

Aitezaz Sikandar Khan

Full - Stack Developer

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Professional Summary

Full-Stack Developer with expertise in building real-time web applications, REST APIs, and containerized services using React, Next.js, Node.js, and MongoDB. Experienced in designing backend services, database schemas, and deployment pipelines with Docker and GitHub Actions. Looking to contribute to a collaborative engineering team as a Full-Stack Developer.

Technical Skills

Languages: JavaScript, TypeScript, HTML5, CSS3, SQL

Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS, Responsive Design

Backend: Node.js, Express.js, RESTful APIs, WebSockets (Socket.io)

Databases: MongoDB, Mongoose ODM, MySQL, Firebase

DevOps: Docker, AWS EC2, Nginx

Tools & Services: Git & Version Control, Postman, Jest, Zod Validation, Stripe API, Cloudinary, Gemini & Groq APIs, Agile

Projects

Collaborative Study Platform | *React, Socket.io, Node.js, Gemini & Groq*

- Developed a real-time virtual classroom utilizing Socket.io to synchronize slide changes, active users, and typing indicators with sub-100ms latency for up to 50 concurrent students.
- Designed a multi-LLM fallback mechanism that automatically redirects requests from Groq (Llama 3) to Google Gemini (1.5-Flash) when rate limits are hit, preventing AI chat interruptions.
- Eliminated audio latency by streaming text-to-speech voice narration directly to the browser using Express chunked transfer encoding headers.

Hospital Management System | *React, Node.js, Express, MongoDB, Gemini API*

- Built a secure, multi-role portal (Admin, Doctor, Patient) with appointment booking, using moment.js validation to prevent scheduling conflicts and Nodemailer for automated email reminders.
- Integrated an AI symptom-analysis assistant via the Gemini API, structuring prompt fallbacks to summarize patient history and generate relevant questions for doctor visits.
- Designed MongoDB aggregation pipelines to calculate monthly appointment distributions and doctor earnings, rendering the data on interactive dashboard charts.

DevOps-Enabled Blog Platform | *React, Node.js, Docker, GitHub Actions, AWS EC2*

- Packaged the React frontend and Node.js backend into multi-stage Docker containers, reducing image sizes via alpine bases to optimize deployment speed.
- Automated the deployment pipeline by writing a GitHub Actions workflow that builds Docker images and redeploys them to AWS EC2 via secure SSH commands.
- Configured Nginx as a reverse proxy to route public traffic, handle backend API requests, and serve compressed assets for reduced page load times.

E-Commerce Platform | *Next.js, TypeScript, Redux, Stripe API, MongoDB*

- Developed a fully typed storefront using Next.js App Router and TypeScript, managing persistent shopping cart states across pages with Redux Toolkit.
- Integrated Stripe Checkout for secure payments, implementing a backend webhook route to save orders to MongoDB and reset customer carts upon payment verification.
- Secured checkout forms and API endpoints against invalid data and malicious inputs using schema-based Zod validation and React Hook Form.

Professional Experience

Freelance Full-Stack Developer

Self-Employed

2025 -- Present

- Developed, deployed, and maintained 3 custom MERN applications for local clients (including a complaint management system, a personal ledger, and a blog app), managing the full cycle from database schema design to hosting.
- Built a complaint tracking system featuring automated email notifications via Nodemailer, media uploads with Cloudinary, and priority-based task queues.
- Secured user authentication and authorization routes using JSON Web Tokens (JWT) and Bcrypt password hashing, ensuring proper endpoint access controls.

Education

University of Peshawar

Peshawar, Pakistan

Bachelor of Science in Computer Science (GPA: 3.60 / 4.00)

2022 -- 2026

- Achievements:** Secured the 95th percentile nationwide in the HEC National Skills Competency Test (NSCT) organized by HEC, PSEB, and P@SHA (April 2026).
- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Software Engineering